DBMS LAB — EXPERIMENT 05

Views and Recursive CTE

Note: These are expected/representative outputs based on the described SQL and sample data, not screenshots from a live MySQL run. Exact formatting and EXPLAIN plans can vary by MySQL version.

Output 1: Department_Salary_Summary view

Dept_ID	Dept_Name	Count	Average	Total	Highest	Lowest
1	Human Resources	5	52,000.00	260,000	60,000	45,000
2	Information Technology	7	87,857.14	615,000	105,000	75,000
3	Finance	6	73,666.67	442,000	90,000	62,000
4	Marketing	6	68,500.00	411,000	80,000	58,000
5	Research and Development	6	94,666.67	568,000	110,000	85,000

Output 2: Employee_Hierarchy view

Emp_ID	Emp_Name	Manager_ID	Manager_Name
1	Aarav Sharma	NULL	NULL
2	Ananya Gupta	NULL	NULL
3	Riya Verma	2	Ananya Gupta
4	Karan Singh	2	Ananya Gupta
6	Rahul Kumar	10	Vivek Gupta
…	…	…	…

Output 3: Recursive reporting chain

The recursive CTE starts from employees with no manager and recursively lists their direct and indirect reports. Row order and depth/path columns depend on the exact SELECT clause in the script. Employees with Manager_ID NULL appear as hierarchy roots.

Safety note

Run this experiment in its own schema or after backing up any existing tables because the script may recreate the experiment tables.